

Optimizing football squad selection: A multi-objective approach to transfer strategy under budget constraints

(DS Area Brown Bag Series)

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Research Overview

Theory / Methods

Time Series Modelling
forecasting, structural breaks, volatility

Quantile Regression
distributional effects and tail behaviour

Spatio-temporal Methods
spatial dependence, dynamic processes

Statistical Learning
prediction, classification, NN

Applications / Case Studies

Environmental Research
pollution, climate

Sports Analytics
football, cricket, tennis

Finance & Risk Analytics
real estate, portfolios, markets

Miscellaneous
business, policy, public health

Outline of today's talk

- 1 Motivation
- 2 Methodology Overview
- 3 Modeling player ratings
- 4 Modeling transfer fees
- 5 Optimization Framework
- 6 Competitive bidding strategy
- 7 Concluding remarks

Collaborators



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Newcastle University

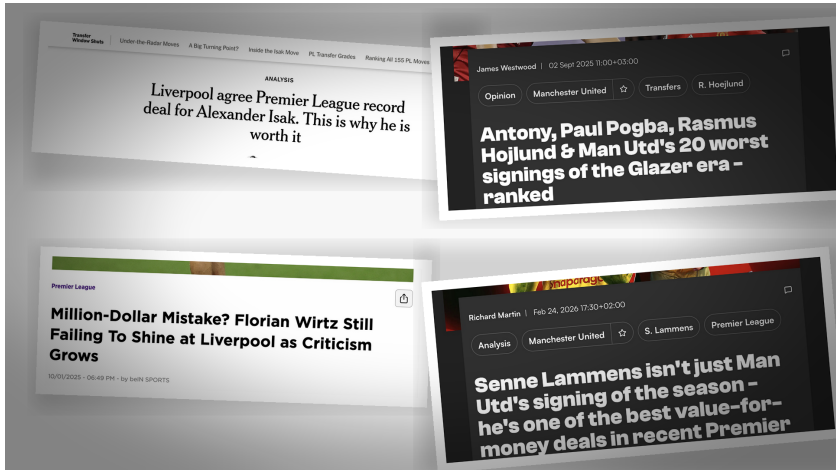


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It's always on the news



Introduction and motivation

- Football transfers are high-stakes, time-bound, and uncertain decisions: clubs buy and sell players to reshape squads.
- The transfer market is not only a pricing task; it is a **decision problem**:
 - predict future contribution (sporting value),
 - predict transfer fees (financial cost),
 - optimize squad composition under constraints.
- Player mobility is high across Europe: average tenure at a club is roughly 1.87–2.56 years.

Auctions, drafts, and transfers

Auctions

- Centralized bidding.
- Budget and time constraints.
- Requires dynamic valuation and strategy.

Drafts

- Sequential allocation
- Requires dynamic planning.
- Multi-stage decision structure.

Transfers (our focus)

- Decentralized negotiations and clauses.
- Multiple objectives: performance, cost, risk.
- Competition can arise simultaneously for the same target player.

What the literature has done

- **Player valuation and fee prediction:** regression, multilevel models, and ML using player traits and performance; some benchmark against crowd valuations. ([Müller et al, 2017](#); [Franceschi et al., 2024](#))
- **Economic view of fees:** bargaining and surplus-sharing perspectives that explain why observed fees may differ from productivity measures. ([Carmichael et al., 1999](#); [Campa, 2022](#))
- **Investment and uncertainty:** recent work frames transfers as risky projects with uncertain sporting and financial returns. ([Tunaru et al., 2005](#); [Follert & Gleißner, 2024](#))

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- **Investment and uncertainty:** recent work frames transfers as risky projects with uncertain sporting and financial returns. ([Tunaru et al., 2005](#); [Follert & Gleißner, 2024](#))
- **Research gap:** studies focus on valuation accuracy for individuals; fewer provide an integrated, constraint-aware decision model.

How this paper contributes

- ① **Two predictive models grounded in club/league structure**
 - next-season rating forecast using linear mixed effects,
 - transfer fee forecast using linear mixed effects on $\log(\text{fee})$.
- ② **A multi-objective optimization framework**
 - jointly balances quality, expected cost, and risk,
 - chance constraint for budget feasibility,
 - operational constraints: squad size, positional composition, retention, profit, transfer-count.
- ③ **Competitive layer (extension)**
 - embeds optimal transfer decisions in an independent private-value bidding setting with reserve to reason about contested targets.

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Workflow

- **Step 1: rating model.** Forecast $\hat{R}_{i_{c_0}, s+1}$ for all candidate players (counterfactual rating at the focal club c_0).
- **Step 2: fee model.** Forecast $\hat{Y}_{i, c \rightarrow c_0, s}$ for buying targets and the expected selling prices for current-squad players.
- **Step 3: optimization.** Choose $x_i \in \{0, 1\}$ (buy/sell decisions) to maximize quality while controlling expected cost and risk, subject to budget and squad constraints.
- **By-product:** Under optimal recommendations, analyze competitive bidding strategy

Key notations

- Teams $\mathcal{C}$, focal team $c_0 \in \mathcal{C}$; players indexed by $i \in [N]$.
- Decision variable $x_i \in \{0, 1\}$ indicates whether player i is selected in the proposed squad.
- Ratings: $R_{ics} \in \mathbb{R}$ for player i , club c , season s .
- Transfer fee for move $c \rightarrow c'$ after season s : $Y_{i,c \rightarrow c',s}$.
- Log-fee: $Z_{i,c \rightarrow c',s} = \log Y_{i,c \rightarrow c',s}$.

Dataset

- Extracted from WhoScored and Transfermarkt
- Data used span a decade: the seasons 2009/10 to 2019/20
- 20403 players from 500 clubs playing in the leagues of 17 different countries from four continents

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Modeling player value via ratings

- Goal: forecast counterfactual rating if player i joins c_0 in season $s + 1$.
- Linear mixed-effects structure:

$$R_{ics} = \mu_{ics} + u_{(c_{s-1} \rightarrow c_s)} + u_{\ell(c_s)}^{\text{cur}} + u_{\ell(c_{s-1})}^{\text{last}} + \varepsilon_{ics}.$$

- Fixed effects in μ_{ics} include: age (quadratic), anthropometrics, position, last-season rating, team context, position-group strength/depth, transfer history, same-team, same-nationality.
- Random effects: origin–destination corridor, current league, last league.

Out-of-sample rating prediction for the focal club

- For candidate player i at club c_0 in season $s + 1$, form features $\mathbf{x}_{i c_0, s+1}$ using information at end of season s :

$$\hat{R}_{i c_0, s+1} = \mathbf{x}_{i c_0, s+1}^\top \hat{\beta} + \hat{u}_{(c_s \rightarrow c_0)} + \hat{u}_{\ell(c_0)}^{\text{cur}} + \hat{u}_{\ell(c_s)}^{\text{last}}.$$

- If corridor $(c_s \rightarrow c_0)$ is unobserved, we shall set $\hat{u}_{(c_s \rightarrow c_0)} = 0$ (marginal forecast).
- These $\hat{R}_{i c_0, s+1}$ are the inputs for squad optimization.

Key insights

- **Age profile is nonlinear:** ratings exhibit diminishing returns, supporting a prime-age focus in transfer planning.
- **Strong persistence:** last season's rating is the dominant predictor.
- **Context matters:** higher team quality and positional strength are strongly associated with higher individual ratings, reinforcing the need for squad-level (not player-by-player) optimization.
- **Depth has benefits:** team depth is positively associated with ratings, motivating position-specific constraints in the optimizer.
- **Mobility penalty signal:** more prior transfers are associated with slightly lower subsequent ratings, suggesting adaptation/instability costs that the optimization can treat as an additional risk.

Results

Type of feature	Variable	Estimate	Standard error	t-score	p-value
General	Intercept	-2.828***	0.140	-20.135	0.000
Player's information	Age (scaled)	0.570***	0.038	15.127	0.000
	Age squared (scaled)	-0.106***	0.007	-15.121	0.000
	Position (defender)	-0.009	0.009	-1.013	0.311
	Position (midfielder)	0.015	0.009	1.655	0.098
	Position (forward)	0.035***	0.007	4.738	0.000
	Height	0.024	0.018	1.297	0.195
	Weight	0.002***	0.000	7.753	0.000
	Last season's rating	0.303***	0.005	65.004	0.000
	Number of times transferred	-0.004**	0.001	-2.867	0.004
	If same team as last season	-0.003	0.003	-0.738	0.461
	Playing in the birth country	-0.007*	0.003	-2.105	0.035
Team's information	Overall quality	0.257***	0.021	12.151	0.000
	Quality in same position	0.723***	0.011	67.059	0.000
	#players in same position	0.004***	0.001	5.393	0.000

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Transfer fee model I

- Transfer fees are strictly positive.
- Heavy right tail in empirical data.
- Linear structure becomes tractable on log scale.
- Define the key response variable as

$$Z_{i,c \rightarrow c',s} = \log Y_{i,c \rightarrow c',s}$$

Transfer fee model II

- Linear mixed effects with buyer/seller heterogeneity:

$$Z_{i,c \rightarrow c',s} = \nu_{i,c \rightarrow c',s} + b_{c'}^{\text{buy}} + b_c^{\text{sell}} + \epsilon_{i,c \rightarrow c',s}.$$

- $\nu_{i,c \rightarrow c',s}$ includes player characteristics, recent performance, career rating, and market/team context.
- Random intercepts capture systematic mark-ups and discounts by buying and selling clubs.
- A similar framework was adopted by [McHale & Holmes \(2023\)](#).

Transfer fee model: mean structure

Mean term $\nu_{i,c \rightarrow c',s}$ is additive in:

- **market trend:** t_s (captures inflation in fees),
- **biographical:** age (quadratic), height, weight, position,
- **quality:** career rating CR_{is} and most recent rating R_{ics} ,
- **recent performance:** matches, goals, assists, shots, passing accuracy, cards, clearances, interceptions,
- **market context:** club depth and club strength, purchasing and selling patterns of the buyer and the seller league.

Transfer fee model: buyer and seller heterogeneity

- Buyer and seller clubs exhibit persistent, unobserved effects:
 - some clubs systematically pay premiums,
 - some clubs systematically extract higher prices as sellers.
- We capture this via random intercepts:

$$b_{c'}^{\text{buy}} \sim \mathcal{N}(0, \sigma_{\text{buy}}^2), \quad b_c^{\text{sell}} \sim \mathcal{N}(0, \sigma_{\text{sell}}^2), \quad \epsilon_{i,c \rightarrow c',s} \sim \mathcal{N}(0, \tau^2).$$

- Interpretation:
 - $b_{c'}^{\text{buy}} > 0$: buyer c' tends to overpay relative to observables,
 - $b_c^{\text{sell}} > 0$: seller c tends to command higher fees.

Transfer fee prediction for planning

- For a prospective move $c \rightarrow c_0$, form the feature vector $\mathbf{w}_{i,c \rightarrow c_0,s}$.
- Predicted log-fee:

$$\hat{Z}_{i,c \rightarrow c_0,s} = \mathbf{w}_{i,c \rightarrow c_0,s}^\top \hat{\boldsymbol{\theta}} + \hat{b}_{c_0}^{\text{buy}} + \hat{b}_c^{\text{sell}}.$$

- If buyer or seller has no support in training data, set the corresponding random effect to 0 (marginal forecast).
- To obtain a fee on the original scale:
 - baseline point forecast: $\hat{Y}_{i,c \rightarrow c_0,s} = \exp(\hat{Z}_{i,c \rightarrow c_0,s})$,
 - log-normal mean correction (if desired): $\hat{\mathbb{E}}(Y) = \exp(\hat{Z} + \frac{1}{2}\hat{\tau}^2)$.
- These predicted fees feed directly into the optimization.

Key insights

- **Market inflation is real:** positive time trend captures systematic fee growth over seasons, so planning needs an inflation adjustment.
- **Lifecycle pricing is concave:** age enters nonlinearly, implying peak pricing around prime years and depreciation thereafter.
- **Roles are different submarkets:** forwards (and to a lesser extent midfielders) command a premium.
- **Long-run quality anchors prices:** career rating is a strong driver of fees, while short-run performance signals refine pricing at the margin.
- **Context calibrates cross-league deals:** league-specific effects shift price levels materially, which is essential when comparing targets across leagues in the squad optimization.

Results I

Type of feature	Variable	Estimate	Standard error	t-score	p-value
General	Intercept	-17.552***	2.156	-8.140	0.000
	Linear trend	0.064***	0.012	5.340	0.000
Player's overall characteristics	Age (scaled)	2.361**	0.743	3.178	0.002
	Age squared (scaled)	-0.668***	0.150	-4.459	0.000
	Position (defender)	-0.323	0.198	-1.635	0.102
	Position (midfielder)	-0.143	0.197	-0.726	0.468
	Position (forward)	0.089	0.167	0.533	0.594
	Height	1.692**	0.520	3.251	0.001
	Weight	0.001	0.004	0.181	0.857
	Career rating	1.601***	0.129	12.392	0.000

Results II

Type of feature	Variable	Estimate	Standard error	t-score	p-value
Player's performance in last season	Rating	-0.373**	0.136	-2.745	0.006
	Game-time	0.018***	0.003	6.036	0.000
	Goals	0.399	0.342	1.166	0.244
	Goal contributions	0.270	0.266	1.015	0.310
	Penalty accuracy	-0.461	0.699	-0.659	0.510
	Shots	0.169***	0.043	3.894	0.000
	Passing accuracy	0.023***	0.005	4.216	0.000
	Cards	0.108	0.156	0.692	0.489
	Clearance	-0.006	0.020	-0.303	0.762
	Interception	-0.061	0.043	-1.401	0.161

Results III

Type of feature	Variable	Estimate	Standard error	<i>t</i> -score	<i>p</i> -value
Leagues and teams in last season	Median selling price (seller league)	0.169***	0.019	8.966	0.000
	Median buying price (buyer league)	0.056***	0.010	5.665	0.000
	#players in same position (seller)	−0.007	0.014	−0.522	0.602
	#players in same position (buyer)	−0.003	0.012	−0.219	0.827
	Quality in same position (seller)	−0.127	0.198	−0.644	0.520
	Quality in same position (buyer)	0.140	0.197	0.712	0.477
	Overall quality (seller)	0.748*	0.302	2.474	0.013
	Overall quality (buyer)	−0.184	0.198	−0.932	0.351

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Optimization: objective and decision variables

- **Decision:** $x_i \in \{0, 1\}$ selects the proposed squad from $\mathcal{P}$.
- **Inputs:** predicted rating R_i and predicted fee Y_i .
- Consider the following key terms:

$$\text{cost} = \sum_{i \in \mathcal{P} \setminus \mathcal{P}_{c_0}} x_i \mathbb{E}(Y_i) + \sum_{i \in \mathcal{P}_{c_0}} (1 - x_i) (\mathbb{E}(Y_i) - r_i),$$

$$\text{risk} = \sqrt{\sum_{i \in \mathcal{P} \setminus \mathcal{P}_{c_0}} x_i \text{Var}(Y_i)}, \quad \text{quality} = \sum_{i \in \mathcal{P}} x_i R_i.$$

- We focus on the objective

$$\mathcal{F} = -(\lambda_1 \text{cost} + \lambda_2 \text{risk}) + \lambda_3 \text{quality}.$$

Budget feasibility via chance constraint

- Total spend must stay within transfer budget $B_{\max}$ with high confidence:

$$\Pr\left(\sum_{i \in \mathcal{P} \setminus \mathcal{P}_{c_0}} x_i Y_i \leq B_{\max}\right) \geq 1 - \alpha.$$

- Deterministic approximation used in the solver:

$$\mathbb{E}\left(\sum_{i \in \mathcal{P} \setminus \mathcal{P}_{c_0}} x_i Y_i\right) + z_\alpha \sqrt{\text{Var}\left(\sum_{i \in \mathcal{P} \setminus \mathcal{P}_{c_0}} x_i Y_i\right)} \leq B_{\max}.$$

- Additional constraints: squad size, position minima/maxima, retention, profit, transfer-count, exposure constraints.

Constraints in practice

Squad and sporting constraints

- total squad size $\leq \overline{k_{\text{tot}}}$
- position-wise minimum
- maximum number of GK
- retain $\geq \underline{k_{\text{retain}}}$ current players
- average age does not increase
- average rating does not decrease

Financial and market constraints

- chance constraint on total spend
- max no of transfers $\leq \overline{k_{\text{transfer}}}$
- min profit from sales $\geq \underline{\text{profit}}$
- brand exposure constraints
- optional rules: must-buy / must-sell / no-sale list

Solving the constrained optimization

Fitness function:

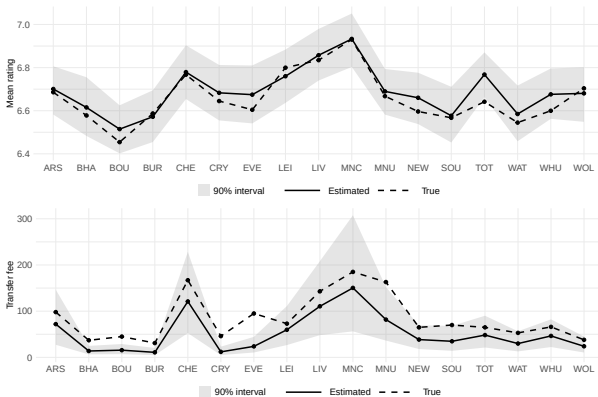
$$\mathcal{F}_\beta(x) = \left[-(\lambda_1 \text{ cost} + \lambda_2 \text{ risk}) + \lambda_3 \text{ quality} \right] - \beta \sum_{k=1}^K \max(0, \text{Constraint}_k),$$

- We use a genetic-algorithm solver with a penalty formulation.
- K equals the number of constraints.
- $\beta \gg 0$ discourages infeasible solutions
- Outputs: the algorithm provides recommended squad, buy list, sell list, and implied cost/risk profile.

Key insights from the algorithm output

- Improved (but not significantly) predicted squad ratings for all but two clubs under consideration.
- Budget-respecting transfer portfolios.
- Identifies:
 - undervalued targets
 - overvalued sell candidates
- Demonstrates portfolio-level decision advantage.

Results: summary



Observed (dashed) versus estimated (solid) club-level means for player ratings (top) and average transfer fees (bottom), with shaded 90% uncertainty bands around the estimated means.

Results: high-valued transfers I

Comparison of recommended (according to our approach, after the 2018/19 season) and recorded transfers for players valued above €20 million who were actually sold.

Team	Player	Position	Expected Price	IQR	Actual Price	Rating
<i>Panel A: Recommended and sold</i>						
Chelsea	Eden Hazard	Forward	208	[58, 243]	121	7.68
Crystal Palace	Aaron Wan-Bissaka	Defender	24	[7, 28]	55	7.52
Leicester City	Harry Maguire	Defender	54	[15, 63]	87	7.08
Manchester City	Fabian Delph	Midfielder	29	[8, 34]	10	7.16
Manchester United	Romelu Lukaku	Forward	80	[22, 93]	74	7.29
Manchester United	Chris Smalling	Defender	31	[9, 36]	Loan	7.24
Southampton	Mario Lemina	Midfielder	32	[9, 38]	Loan	6.93
Tottenham Hotspur	Kieran Trippier	Defender	44	[12, 51]	22	7.04
West Ham United	Andy Carroll	Forward	26	[7, 30]	Free	6.69

Results: high-valued transfers II

Comparison of recommended (according to our approach) and recorded transfers for players valued above €20 million who were not sold in that year. The year in the parentheses indicates when the player was eventually transferred, 'Free': free agent.

Team	Player	Position	Expected Price	IQR	Actual Price	Rating
<i>Panel B: Recommended but not sold</i>						
Liverpool	Sadio Mane	Forward	76	[21, 89]	32 (2022)	7.27
Liverpool	Andrew Robertson	Defender	28	[8, 32]	Free (2026)	7.02
Liverpool	Jordan Henderson	Midfielder	23	[7, 27]	14 (2023)	6.95
Manchester City	Bernardo Silva	Midfielder	59	[16, 69]	Free (2026)	6.79
Manchester City	Riyad Mahrez	Forward	53	[15, 61]	35 (2023)	7.28
Southampton	Pierre Emile Hojbjerg	Midfielder	27	[7, 31]	17 (2020)	6.61

Results: high-valued transfers III

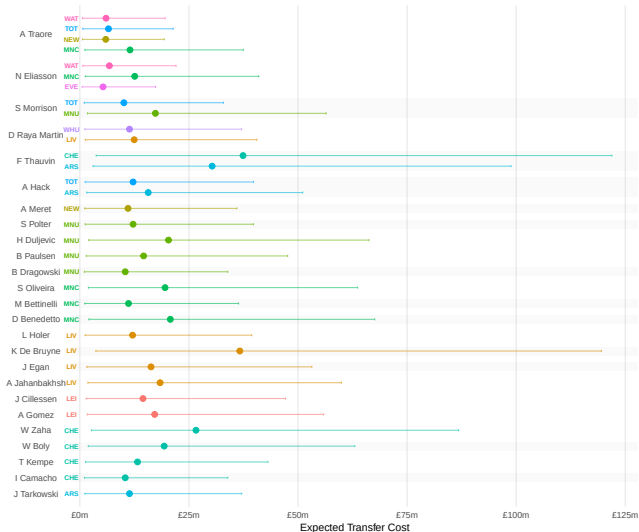
Comparison of expected and recorded transfer fees for players not recommended but sold in 2019. 'Free': free agent, 'Loan': loaned out.

Team	Player	Position	Expected Price	IQR	Actual Price	Rating
<i>Panel C: Not recommended but sold</i>						
Arsenal	Alex Iwobi	Midfielder	24	[7, 28]	30	6.88
Arsenal	Aaron Ramsey	Midfielder	20	[6, 24]	Free	7.27
Chelsea	David Luiz	Defender	43	[12, 50]	9	6.85
Liverpool	Daniel Sturridge	Forward	30	[8, 34]	Free	6.76
Manchester City	Danilo	Defender	74	[21, 86]	37	6.91
Manchester United	Alexis Sanchez	Forward	49	[14, 58]	Loan	7.55
Manchester United	Ander Herrera	Midfielder	35	[10, 40]	Free	6.60
Manchester United	Marouane Fellaini	Midfielder	28	[8, 33]	7	6.94
Newcastle United	Ayoze Perez	Forward	26	[7, 30]	33	6.74
Southampton	Charlie Austin	Forward	25	[7, 29]	4	6.53
Southampton	Matt Targett	Defender	24	[7, 28]	16	6.96
Tottenham Hotspur	Mousa Dembele	Midfielder	28	[8, 33]	5	6.90
West Ham United	Marko Arnautovic	Forward	32	[9, 37]	25	7.07

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Players with (recommended) multiple bidders



Scope and motivation I

- In many transfers, multiple clubs target the same player, so the final fee reflects **strategic competition**, not just predicted market value.
- Clubs have **asymmetric willingness-to-pay** (financial strength, sporting objectives), motivating an asymmetric first-price auction framework.
- Sellers may reject the highest bid due to uncertainty and negotiation, so we model a **random reserve price** and allow a **no-sale** outcome.
- Football-specific frictions matter: inter-club relationships may affect deals, captured through a **bid-dependent acceptance** function.

Scope and motivation II

- Goal: translate **contested target** situations into an equilibrium-based tool for expected price, win probability, and negotiation dynamics.
- We take inspirations from the works of [Lebrun \(1999\)](#); [Li & Perrigne \(2003\)](#); [Kotowski \(2018\)](#).

Bidding setting for a contested target

- When multiple clubs target the same player, fees reflect **strategic bidding** and **asymmetric budgets**.
- Model the situation as an **asymmetric first-price auction** with independent private valuations:

$$\log S_c \sim \mathcal{N}(\mu_c, \sigma_c^2), \quad c \in \mathcal{C}^i.$$

- Each club uses an increasing bid function $b = \kappa_c(s)$ (inverse $\psi_c = \kappa_c^{-1}$).
- Goal: characterize equilibrium bids and translate them into win probabilities and expected fees for planning.

Seller behaviour: random reserve and affinity

- Seller uncertainty is captured via a **random reserve price**:

$$\log \rho \sim \mathcal{N}(\mu_{c_0}, \sigma_{c_0}^2), \quad H(b) = \Pr(\rho \leq b).$$

- Club-specific deal frictions are captured via **bid-dependent acceptance** $p_c(b) \in [0, 1]$.
- Buyout clause is modelled by truncation at v_{thresh} :

$$\hat{H}(b) = \min \left\{ \frac{H(b)}{H(v_{\text{thresh}})}, 1 \right\}, \quad \hat{p}_c(b) = \min \left\{ \frac{p_c(b)}{p_c(v_{\text{thresh}})}, 1 \right\}.$$

- This admits **no-sale** outcomes and captures rivalry/relationship effects.

Equilibrium characterization (main result)

- Probability that club c outbids rivals at bid b :

$$G_c(b) = \prod_{j \neq c} F_j(\psi_j(b)).$$

- Probability of a completed sale to club c at bid b :

$$q_c(b) = \hat{p}_c(b) \hat{H}(b) G_c(b).$$

- Expected utility for valuation s :

$$U_c(b; s) = (s - b) q_c(b).$$

- Bayes–Nash equilibrium yields an FOC and an ODE system for $\psi_c(b)$, enabling **numerical solution** of bidding strategies under asymmetry, random reserve, and affinity.

Theoretical results

Proposition 1

Suppose $p_c(\cdot)$ and $\psi_c(\cdot)$ are continuously differentiable function on $(0, v_{\text{thresh}}]$. Then, in an interior pure-strategy equilibrium under risk neutrality,

$$\frac{1}{s-b} = \frac{p'_c(b)}{p_c(b)} + \frac{h(b)}{H(b)} + \sum_{j \neq c} \frac{f_j(\psi_j(b))}{F_j(\psi_j(b))} \psi'_j(b).$$

Proposition 2

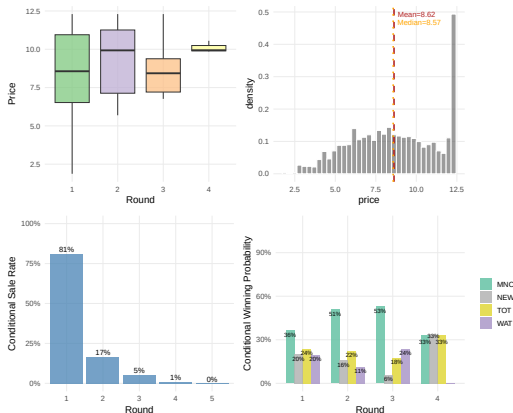
Suppose $p(\cdot)$ and $\psi_c(\cdot)$ are continuously differentiable in the interval $[b_{\min}, v_{\text{thresh}}]$ with $p(b), \psi_c(b) > 0$. Then, we get the following system of ODEs, for all c ,

$$\begin{aligned} \psi'_c(b) = \frac{1}{(C-1)} \frac{F_c(\psi_c(b))}{f_c(\psi_c(b))} & \left[\sum_{j \neq c} \left(\frac{1}{\psi_j(b) - b} - \frac{p'_j(b)}{p_j(b)} \right) \right. \\ & \left. - (C-2) \left(\frac{1}{\psi_c(b) - b} - \frac{p'_c(b)}{p_c(b)} \right) - \frac{h(b)}{H(b)} \right]. \end{aligned}$$

Extension: Bidding in multiple rounds

- Negotiation is modeled as a multiple-round game with the possibility of no-sale.
- In each round, clubs bid, the seller observes the highest bid, then accepts or rejects.
- A rejection updates beliefs about the reserve price, so later rounds become more competitive.
- Buyers update participation by truncating to types that could bid above the last rejected maximum.
- Each round produces an updated equilibrium (solved numerically) with a small bid-gap for stability.

Case study of Adama Traore



(top-left) round-wise price distribution; (top-right) overall price distribution; (bottom-left) round-wise conditional sale rates; (bottom-right) round-wise win % for different teams

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Key takeaways

- A transfer window can be planned as a **multi-objective** decision problem: quality, expected cost, and risk.
- Mixed-effects models give **club- and league-aware** forecasts for both ratings and fees.
- The optimization layer converts forecasts into **actionable buy/sell** recommendations under realistic constraints.
- Competitive pressure for contested targets can be incorporated to refine expected cost and success probability.

Future scope

- Add richer uncertainty: injuries, minutes forecasts, and scenario-based budget stress tests.
- Extend constraints: wages, amortization, registration rules, and multi-window planning.
- Improve price modeling with contract details and deal structure (add-ons, sell-on clauses).
- Calibrate the bidding layer using observed rival interest and seller affinity signals.
- Explore similar framework under the auction mechanism, and work with IPL data.

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